Fraudulent Transaction Detection for Digital Money Transfer

Project Workflow

Step 1: Data Collection & Profiling (Data examined in its native state)

- Analyze distributions, identify missingness patterns, and
- establish baseline fraud prevalence across the dataset.

Step 2: Data Preparation (Data is cleaned)

- Clean data

Step 3: Standardized and Engineer the Data

- Standardize data and
- Engineer features including velocity metrics, mismatch indicators, device fingerprints, and temporal patterns.

Step 4: Exploratory Data Analysis (Data is Explored)

- Uncover patterns by channel, geography, and time
- Identify anomalies distinguishing fraudulent from legitimate behavior.

Step 5: Model Development (Model is developed, Tuned and compared)

- Progress from Logistic Regression and Random Forest baselines to advanced XGBoost and LightGBM models
- Tune hyperparameters and
- Compare performance using confusion matrix, recall, precision, F1-score, and ROC-AUC metrics.

Step 6: Validation & Explainability (Model is validated and interpreted)

- Conduct rigorous holdout evaluation
- Generate SHAP values to provide transaction-level reasoning and build trust in model decisions.

Step 7: Deployment (Model is deployed)

- Build FastAPI /score service endpoint
- containerize with Docker for seamless portability across environments.

Step 8: Monitoring & Improvement (Model is monitored)

- Implement Evidently-based drift detection;
- Establish quarterly retraining playbook to maintain model accuracy as fraud patterns evolve.